In [1]:
```python
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

In [2]:
```python
df = sns.load_dataset('titanic')
df.head()
```

Out[2]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adul
0	0	3	male	22.0	1	0	7.2500	S	Third	man	
1	1	1	female	38.0	1	0	71.2833	C	First	woman	
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	
3	1	1	female	35.0	1	0	53.1000	S	First	woman	
4	0	3	male	35.0	0	0	8.0500	S	Third	man	

In [3]:
```python
df.info()
df.describe()
```

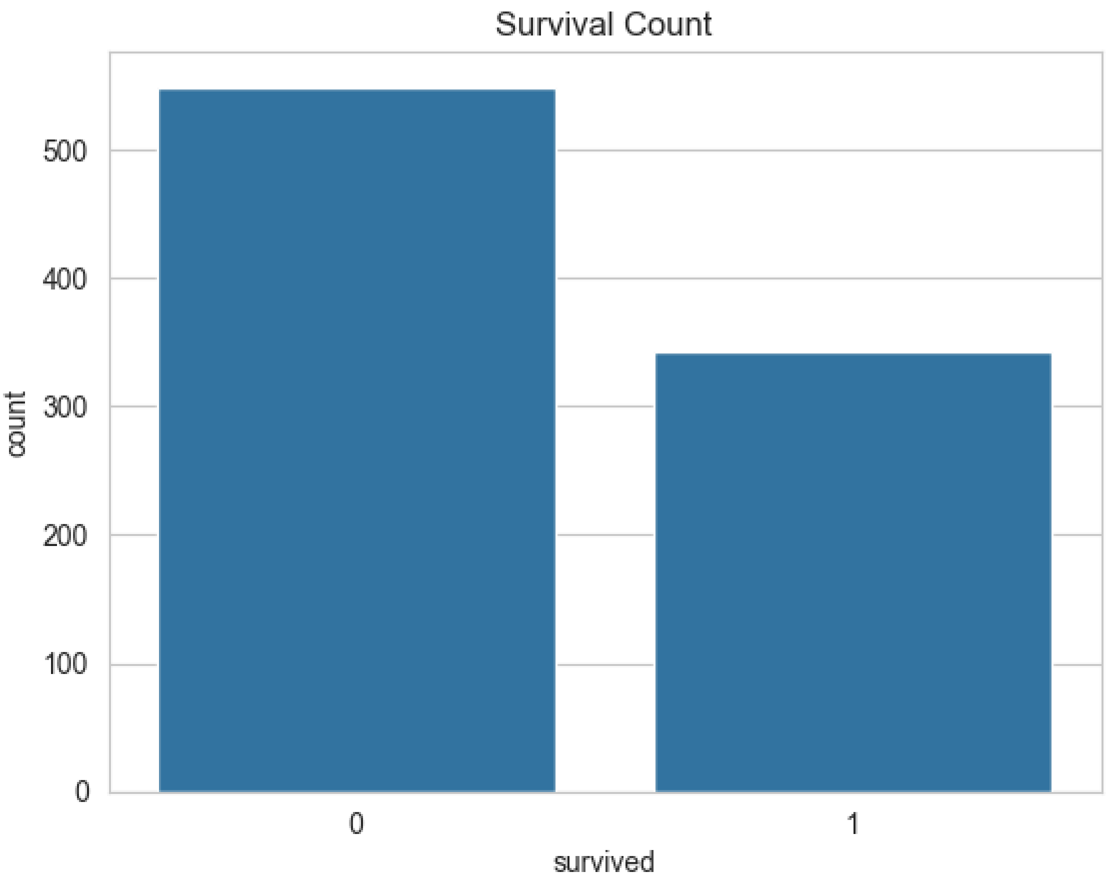
```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    str
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    str
 8   class        891 non-null    category
 9   who          891 non-null    str
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    category
 12  embark_town  889 non-null    str
 13  alive        891 non-null    str
 14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), str(5)
memory usage: 80.7 KB
```

Out[3]:

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

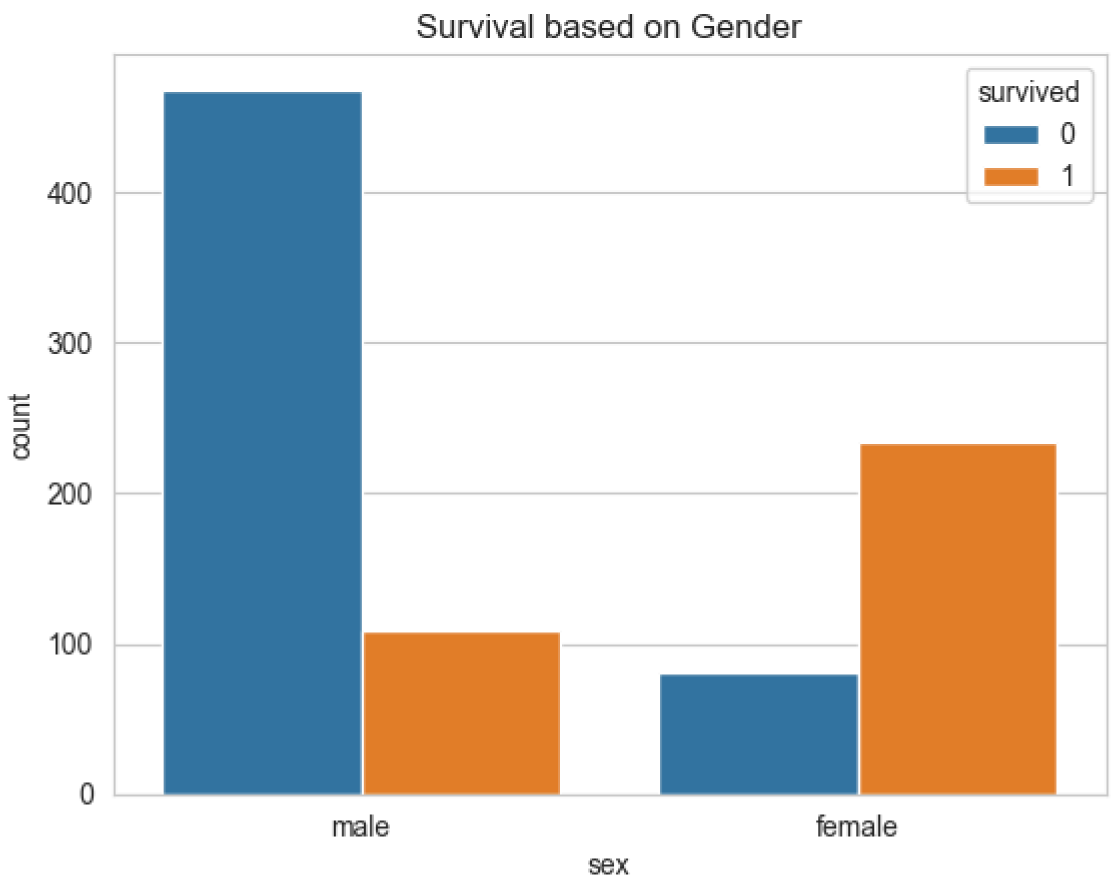
In [7]:

```
sns.countplot(x='survived', data=df)
plt.title("Survival Count")
plt.show()
```

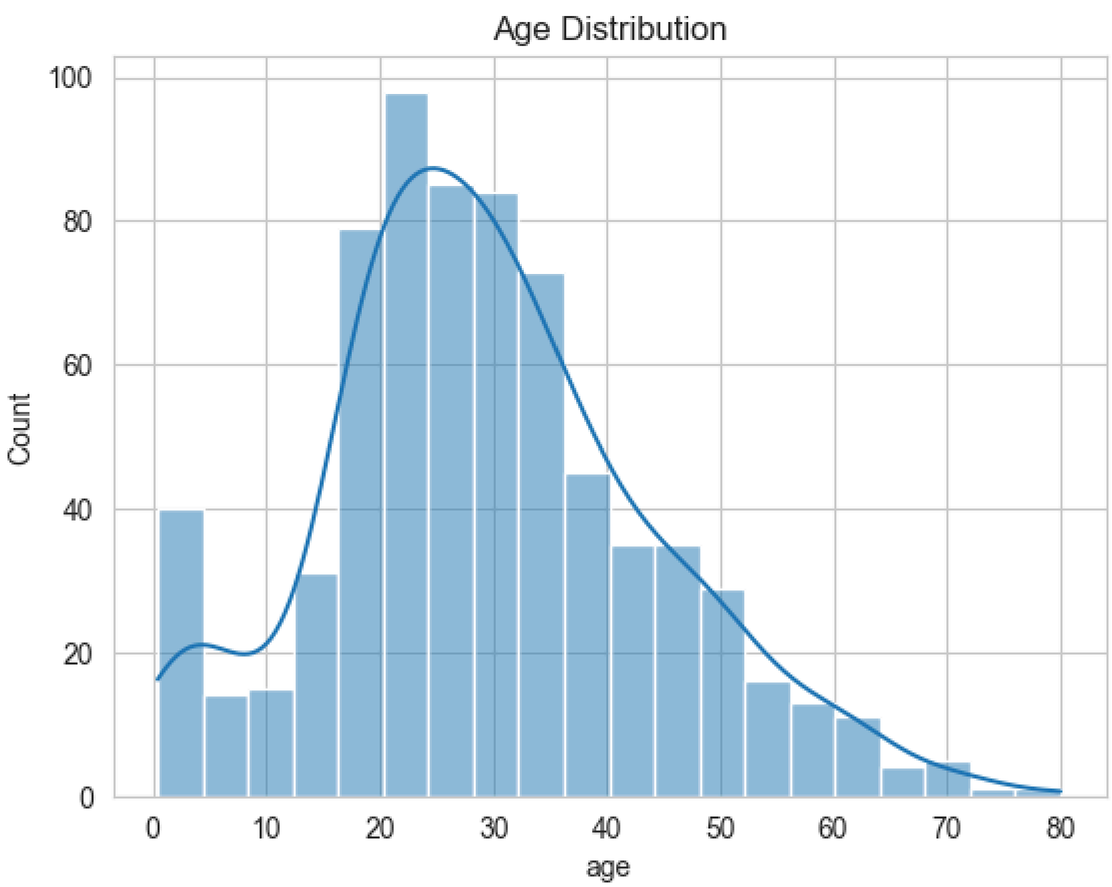


In [8]:

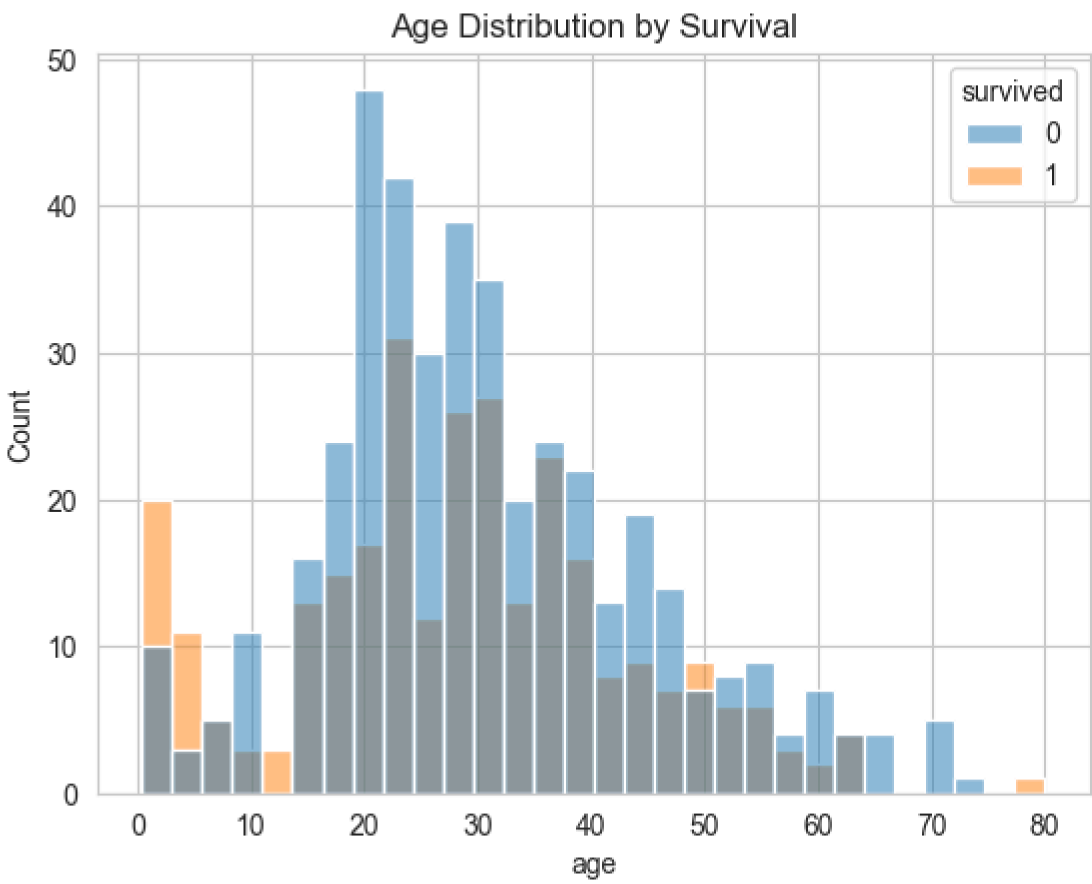
```
sns.countplot(x='sex', hue='survived', data=df)
plt.title("Survival based on Gender")
plt.show()
```



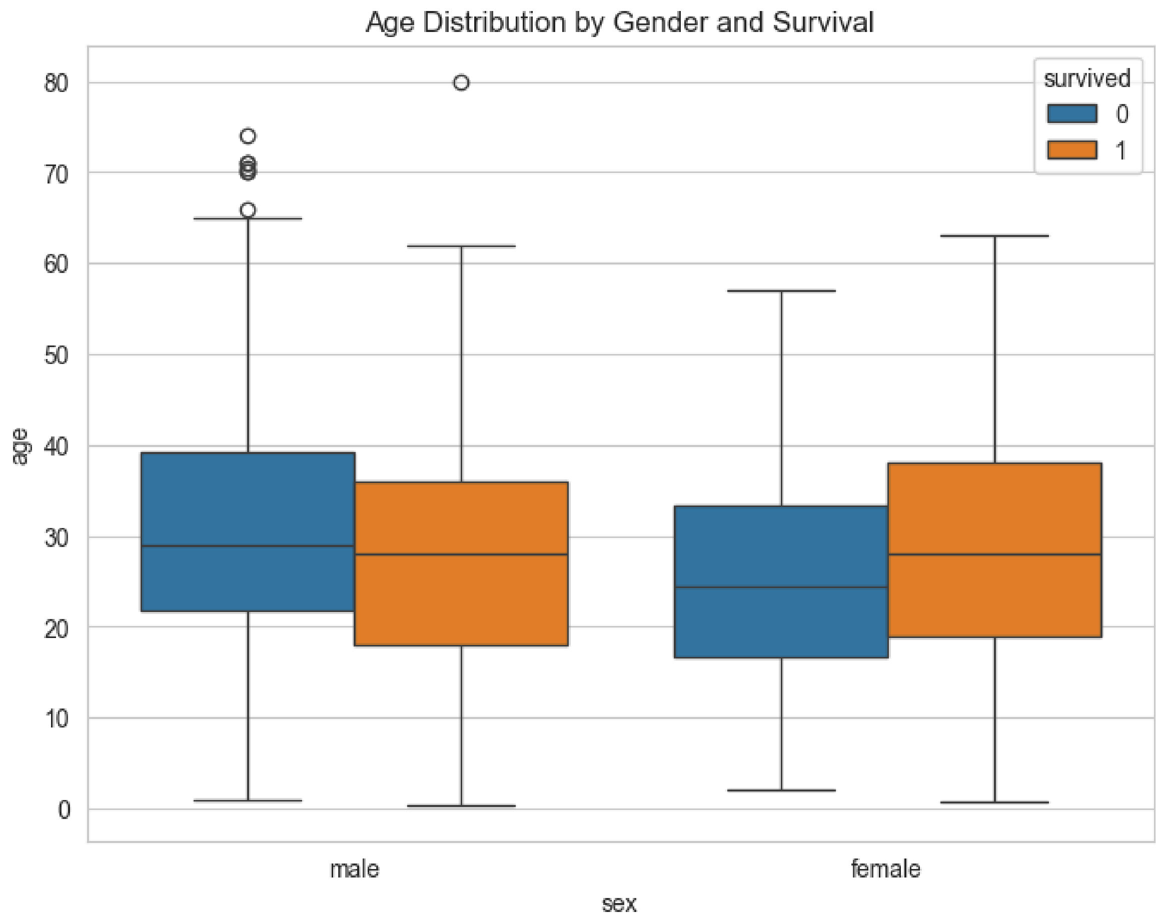
In [9]:
```python
sns.histplot(df['age'], kde=True)
plt.title("Age Distribution")
plt.show()
```



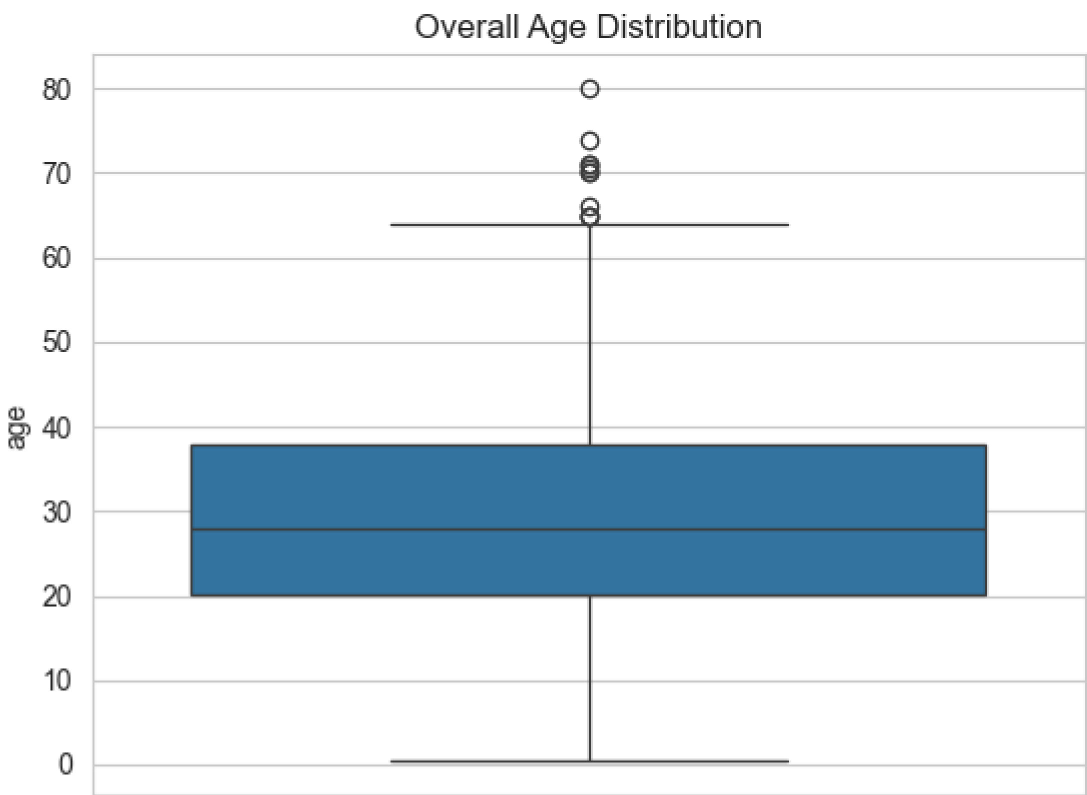
In [10]:
```
sns.histplot(data=df, x='age', hue='survived', bins=30)
plt.title("Age Distribution by Survival")
plt.show()
```



In [11]:
```
plt.figure(figsize=(8,6))
sns.boxplot(x='sex', y='age', hue='survived', data=df)
plt.title("Age Distribution by Gender and Survival")
plt.show()
```

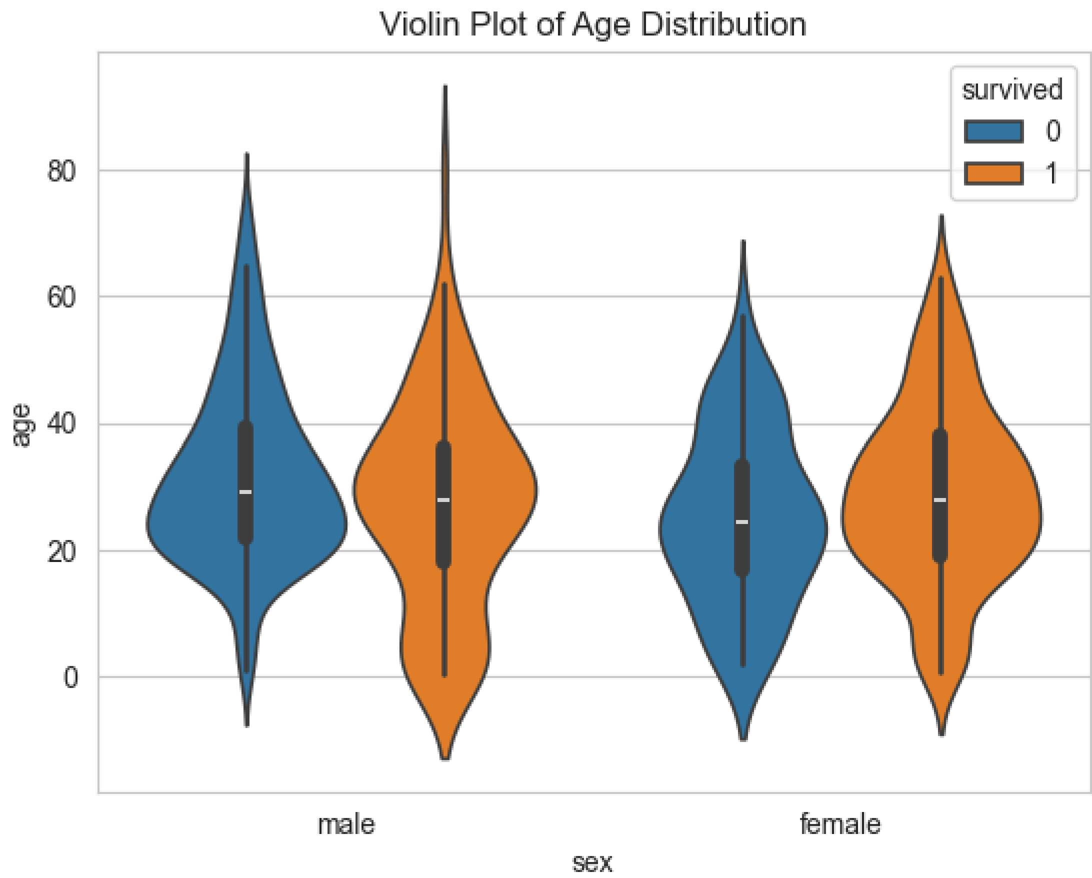


In [12]:
```
sns.boxplot(y='age', data=df)
plt.title("Overall Age Distribution")
plt.show()
```

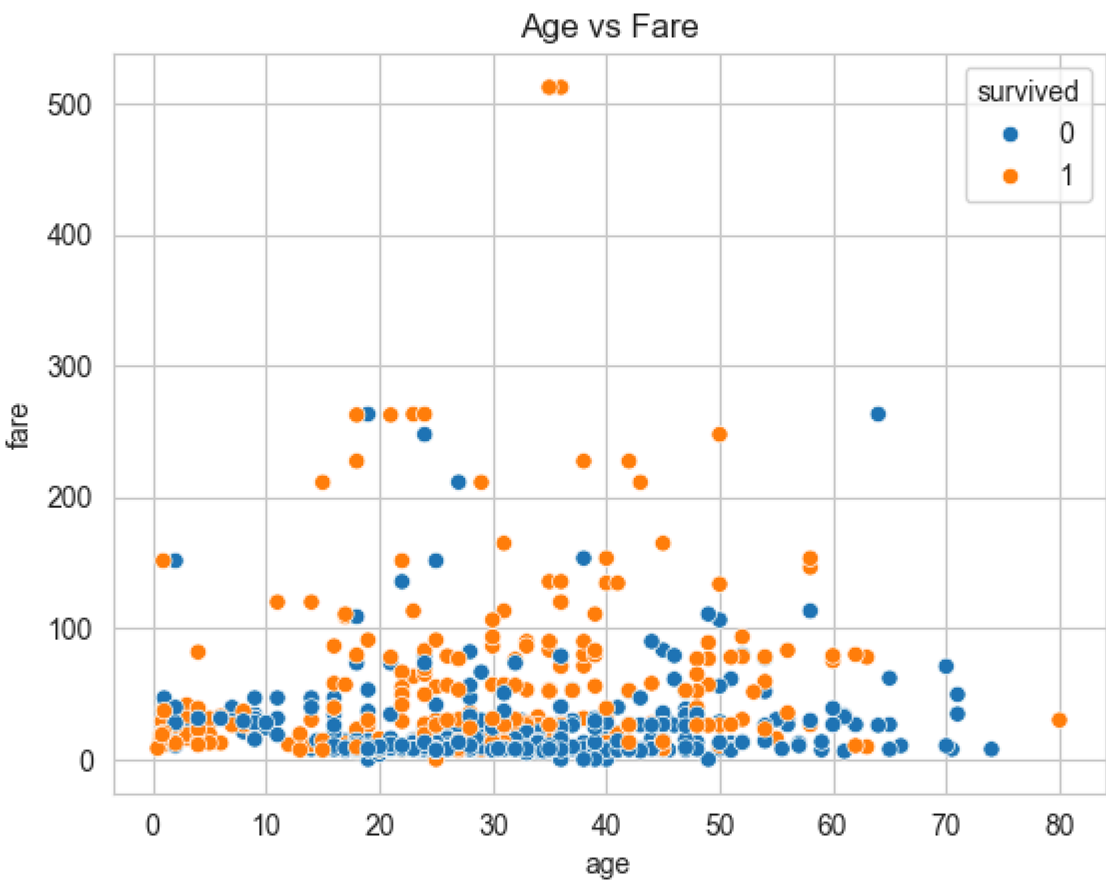


In [13]:
```
sns.violinplot(x='sex', y='age', hue='survived', data=df)
plt.title("Violin Plot of Age Distribution")
```

```
plt.show()
```



In [14]:
```
sns.scatterplot(x='age', y='fare', hue='survived', data=df)
plt.title("Age vs Fare")
plt.show()
```



```
In [ ]:
```